

Thai *Tough*-Constructions and the *Tough* Signature

Rotsuprit Saengthong and John Gluckman

September 14, 2026

Abstract

We investigate the *tough*-construction in Thai, focusing on two exceptional properties. First, Thai *tough*-constructions may involve a potentially unconstrained dependency, and second, the gap in the embedded clause can be in the subject position. While these facts superficially support a heterogeneous view of *tough*-constructions across the world's languages, on closer inspection, we come to a different conclusion. We argue that unconstrained dependencies and subject gaps are not a function of Thai *tough*-constructions, but instead reflect the fact that Thai has an undiagnosed broad subject position. Once controlled for, Thai *tough*-constructions look surprisingly like other well-documented cases. Ultimately, we are concerned not with variation in *tough*-constructions, but rather crosslinguistic similarities.

[*tough*-construction, Thai syntax, broad subjects, narrow subjects, crosslinguistic variation]

1 Introduction

Crosslinguistic research on the *tough*-construction has revealed a number of interesting points of variation; it is an empirical fact that the *tough*-construction has different manifestations across the world's languages (Comrie and Matthews, 1990; Havu and Höglund, 2015; Tayalati et al., 2020). In this paper, we will approach the question of crosslinguistic variation in the *tough*-construction from the perspective of Thai. Our starting point is the alternation in (1).¹

- (1) a. *man yâak [thîi cà? tàt tônmáy tôn nîi]*
EXPL difficult COMP IRR cut tree CLF this
'It is difficult to cut this tree'

¹We use the following abbreviations.

- | | | |
|---------------------------|---------------------|---------------------------|
| • ACC: accusative | • GEN: genitive | • REFL: reflexive |
| • ASP: aspect | • IRR: irrealis | • RSM: resumptive pronoun |
| • CLF: classifier | • NEG: negation | • SG/PL: singular/plural |
| • COMP: complementizer | • NOM: nominative | • TOP: topic |
| • COP: copula | • PART: particle | |
| • DU: dual | • PASS: passive | |
| • EXPL: expletive | • PERF: perfective | |
| • F/M: feminine/masculine | • PROG: progressive | |

When citing from other sources we use the glossing conventions and translations from those sources. Unless otherwise cited, the data come from the first author who is an L1 speaker of Thai.

- b. [tôn máy tôn nĩ]₁ yâak ??([thũi cà? tât ____₁])
 tree CLF this difficult COMP IRR cut
 ‘This tree is difficult to cut.’

In many respects, the data in (1) fit the crosslinguistic pattern of *tough*-constructions: there is an alternation between an expletive and referential subject, suggesting “thematic displacement” akin to subject-to-subject raising, and the antecedent-gap relationship does not display connectivity effects, e.g., reconstruction. However, Thai *tough*-constructions have a couple of striking properties that are not characteristic of more well-studied *tough*-constructions (e.g., English and Romance languages). It has been observed that the Thai *tough*-construction allows entirely unbounded, island-violating, dependencies, like in (2a) (cf, Hattori 2023). Additionally, the gap in the nonfinite clause can be a subject, (2b).

- (2) a. rót khan nĩ₁ yâak mâak thũi Jane cà? sǎcay [CP phrǎ? Sam sũu ____₁]
 car CLF this difficult very COMP Jane IRR upset because Sam buy
 ‘*This car is very difficult for Jane to be upset because he bought.’
 b. [nákrian phũaknĩ]₁ yâak [CP thũi ____₁ cà? thũuk thamthôot]
 student CLF.PL difficult COMP IRR PASS punish
 ‘*These students are difficult to be punished.’

At first blush, the Thai *tough*-construction data suggest that there is a great deal of heterogeneity in the *tough*-construction crosslinguistically, adding to the variation observed in Comrie and Matthews (1990), Havu and Höglund (2015), and Tayalati et al. (2020). The implication is that we would need to devise theoretical tools that are flexible enough to admit languages like Thai, but can be restricted to derive languages like English, which has a more “constrained” *tough*-construction. Indeed, this is the approach of Takita et al. (2022) who examine similar data in Japanese.

However, upon closer inspection, we actually believe that the Thai data point to a different conclusion. Rather than expanding our understanding of *differences* across *tough*-constructions in the world’s languages, we believe that the data actually point to *commonalities* among *tough*-constructions. We show that the *tough*-construction in Thai *does* involve a dependency constrained by movement, and the gap *cannot* be a subject. The examples in (2) represent a different beast: the subjects are broad (or major) subjects, which instantiate a well-known kind of non-movement long-distance dependency, one which involves binding of a (sometimes null) pronoun. Roughly, (2a) is closer to “This car is such that it is very difficult for Sam’s father to be upset because he bought it.” By extension, we will show that (1b) has two parses, one of which is closer to “This tree is such that it is difficult to cut it.” More importantly, once we control for the possibility of a broad subject in (1b), we find that Thai’s *tough*-construction ends up looking remarkably similar to English—and indeed all other known *tough*-constructions. We are thus ultimately concerned with what specific properties constitute the “*tough*-signature”. What are the universal properties of the *tough*-construction? We propose a core set of four properties that define a universal *tough*-signature, which we will outline in section 6.

A by-product of our investigation is a more articulated understanding of Thai “subject” positions. Following work in Cardinaletti (1990, 2004); Heycock and Doron (2003) among others, we motivate the existence of two distinct subject positions in Thai (following preliminary observations in Hoonchamlong 1991; Iwasaki and Ingkaphirom 2005 among others).

Our paper is structured as follows. We will first cover in section 2 the relevant properties of the *tough*-construction in Thai. In section 3 we will review diagnostics that distinguish topics, broad subjects, and narrow subjects in Thai. We will then apply some of those diagnostics to the *tough*-construction in section 4, showing that once we control for higher functional material, Thai’s *tough*-construction does not allow unconstrained long-distance dependencies or subject gaps. In subsection 4.4 we will summarize our findings

and discuss two other languages with a similar state of affairs. Section 5 presents our analysis of the Thai facts. Finally, in section 6 we will address the universals of the *tough*-construction and conclude.

2 Properties of the Thai *tough* construction

Thai has a *tough*-construction. Within the main clause, there is an alternation between an expletive subject, *man* ‘it’, and a non-expletive subject co-indexed with a gap within the embedded clause (3), repeated from (1).²

- (3) a. *man yâak [thîi cà? tàt tônmáy tôn nûi]*
 it difficult COMP IRR cut tree CLF this
 ‘It is difficult to cut this tree’
 b. *[tônmáy tôn nûi]₁ yâak ??([thîi cà? tàt ____₁])*
 tree CLF this difficult COMP IRR cut
 ‘This tree is difficult to cut.’

At first blush, Thai *tough*-constructions fit what is observed in *tough*-constructions in other languages. Most importantly, the example in (3b) involves a thematic displacement. Without the associated clause headed by *thîi*, this sentence cannot be given an interpretation because the subject lacks a thematic role. The same property is found with the “usual” class of adjectives that appear in the *tough*-construction (cf, Gluckman 2021).

- (4) *[năŋsǎn lêm nûi]₁ ŋây/yâak/sămkhan [thîi cà? ?àan ____₁ kòŋ sòp]*
 book CLF this easy/difficult/important COMP IRR read before exam
 ‘This book is easy/difficult/important to read before the exam.’

Also similar to other cases of *tough*-constructions is the observation that the clause headed by *thîi* in Thai is necessarily nonfinite in the *tough*-construction (cf, Comrie and Matthews 1990). This is slightly more difficult to demonstrate in Thai because the complementizer *thîi* can in principle head finite clauses—as we will discuss later. However, in the *tough*-construction the associated clause is diagnosably nonfinite, using tests from Hoonchamlong (1991). The evidence supporting this claim comes from the fact that unlike finite clauses as in (5), embedded clauses in the *tough*-construction are incapable of hosting modals and auxiliaries; instead, the irrealis morpheme *cà?* is obligatory.

²Our paper will be concerned with adjectival *tough*-predicates, but Thai has a “family” of related constructions, like English, including nominal predicates, and verbal predicates like *cost* and *take-TIME* (Gluckman, 2022). Though we will not discuss this data in detail, nothing we say below is contradicted by anything we omit. Note that, according to Hattori (2023), there is an alternative way to form a *tough*-construction in Thai, where the *tough*-predicate appears verb-finally, (i).

- (i) *năŋsǎn nûi ?àan yâak*
 book this read difficult

‘This book is difficult to read ____.’

adapted from Hattori (2023, 33)

- (ii) *tônmáy tôn nûi tàt mûawaan / dūay rûay*
 tree CLF this cut yesterday / with saw
 ‘*This tree cut yesterday/with a saw.’

In fact, we believe that (i) is not a *tough*-construction, but rather, an example of *middle voice* (ii) (which is known to overlap with the *tough*-construction in some respects; Massam 1987; Kim 1995; Kusayama 1998). One reason for thinking this is a middle is that it is compatible with non-*tough* predicates/phrases in this post-verbal position, as shown in (ii). Note that, unlike *tough*-constructions, Thai middle constructions are not restricted to nonfinite or irrealis clauses.

- (5) a. *Sam kamlaj cà? tət tōnmáy tōn nūi*
 Sam PROG IRR cut tree CLF this
 ‘Sam is cutting this tree.’
 b. *Sam khəy tət tōnmáy tōn nūi*
 Sam PERF cut tree CLF this
 ‘Sam cut this tree.’
- (6) a. **[tōnmáy tōn nūi]₁ yāak [thūi (Sam) kamlaj cà? tət ____₁]*
 tree CLF this difficult COMP Sam PROG IRR cut
 Intended: ‘*This tree is difficult that Sam be cutting.’
 b. **[tōnmáy tōn nūi]₁ ḡāy [thūi (Sam) khəy tət ____₁]*
 tree CLF this easy COMP Sam PERF cut
 Intended: ‘*This tree is easy that Sam cut.’³

This observation matches what Singhapreecha (2010, 2019) finds for complement clauses of Subject Control, whose embedded clause contains *cà?* as in (7a, 7b) or *thūi cà?* as in (7c). She ultimately concludes that both variants subcategorize for a non-finite clause.⁴

- (7) a. *Somsak jàak (cà?) thamḡaan naj UN*
 Somsak want IRR work in UN
 ‘Somsak wants to work for the United Nations’
 b. *Somsak jàak *khuḡ/*kamlaj (cà?) thamḡaan naj UN*
 Somsak want PERF/PROG IRR work in UN
 ‘Somsak wants to have worked/be working for the United Nations’
 c. *Somsak jàak thūi (cà?) thamḡaan naj UN*
 Somsak want COMP IRR work in UN
 ‘Somsak wants to work for the United Nations’

(Singhapreecha, 2019, 28)

A final similarity with other languages is that the subject in Thai *tough*-constructions resists reconstructed readings (Epstein, 1989; Fleisher, 2013; Poole et al., 2017). We show this with variable binding in (8) and scope in (9a) and (9b). In (8) a variable in the subject cannot be bound by a quantifier inside of the nonfinite clause. (9a) demonstrates a reconstructed reading of the quantifier is not possible, as it cannot have the same meaning as (9b). (See discussion in Fleisher 2013 in particular.)

- (8) *[krà?pǎw khǒḡ khǎw_{1/*2}]₃ yāak thūi nákrīan thúk khon₂ cà? chǒp ____₃*
 backpack of 3SG difficult COMP student every CLF IRR like
 ‘Her_{1/*2} backpack is difficult for every student₂ to like ____.’
- (9) a. *[nǎḡsǎn sùanyàḡ]₁ yāak thūi Sam cà? ḡàan ____₁*
 book most difficult COMP Sam IRR read
 ‘Most books are difficult for Sam to read.’

³We note incidentally there is no ambiguity about whether there can be an overt subject inside of the nonfinite clause. This topic is debated in English due to the fact that the following *for*-phrase is potentially ambiguous between a PP parse and complementizer parse. Depending on which parse is argued to be possible, the result has implications for the antecedent-gap chain in the *tough*-construction. In Thai, since *thūi* must be a complementizer here—it is not a preposition—the following DP is obligatorily inside of the nonfinite clause. Full (nonfinite) CPs are possible in the *tough*-construction in Thai; this does not resolve the question in English.

⁴See Singhapreecha (2010, 2019) for discussion of the difference in meaning when *cà?* is presence or absent. In the *tough*-construction *cà?* is obligatory.

- b. $\neq$ *man yâak thîi Sam cà? ?àan năjsŋ sàanyà*
 EXPL difficult COMP Sam IRR read book most
 ‘It is difficult for Sam to read most books.’

We thus find that the Thai *tough*-construction exhibits (i) a thematic displacement, (ii) out of the nonfinite clause, (iii) which does not permit reconstruction. This set of properties makes the Thai *tough*-construction superficially similar to nearly all other *tough*-constructions.⁵

Despite these parallels, the Thai *tough*-construction diverges significantly in two respects. First, the Thai *tough*-construction is island-insensitive. The antecedent-gap chain can cross all known barriers for movement in Thai.⁶

(10) *Adjunct island*

- [*rót khan nŋ*]₁ *yâak mâak thîi Jane cà? sŋacay* [_{CP} *phrŋ? Sam sŋ* ____₁]
 car CLF this difficult very COMP Jane IRR upset because Sam buy
 ‘*This car is very difficult for Jane to be upset because he bought.’

(11) *Sentential subject*

- [*rót khan nŋ*]₁ *yâak mâak thîi cà? khŋt wâa* [_{CP} *thîi Sam cà? khăy* ____₁] *pen khàaw*
 car CLF this difficult very COMP IRR think COMP COMP Sam IRR sell COP news
dii
 good
 ‘*This car is very difficult to think that that Sam will sell is good news.’

The facts therefore point towards an analysis of the Thai *tough*-construction in which there is no movement dependency—a surprising and typologically rare pattern. (We return in subsection 4.4 to two other reported cases in Japanese and Modern Standard Arabic.)

The second surprising fact about the *tough*-construction in Thai is that the gap can occupy any position in the embedded clause—including the subject position of the nonfinite clause.

- (12) a. [*tôn máy tôn nŋ*]₁ *yâak* [*thîi* ____₁ *cà? thŋuk tât*]
 tree CLF this difficult COMP IRR PASS cut
 ‘*This tree is difficult to be cut.
 b. *dekdek*₁ *yâak* [*thîi* ____₁ *cà? khâwca* *phaasăa angkrît*]
 children difficult COMP IRR understand language English
 ‘*Children are difficult to understand English.’

The subject/non-subject asymmetry in *tough*-constructions is remarkably robust in other languages; it is the core distinction between *tough*-constructions and subject-to-subject raising, and it is the core similarity with other so-called missing object constructions (Dalrymple and King, 2000; Levine and Hukari, 2006) and complement object deletions (Lasnik and Fiengo, 1974), which constitute a family of related phenomena.

⁵To be clear, there is mixed behavior cross-linguistically with respect to reconstruction. Some languages, like English resist reconstruction entirely, while others show (limited) reconstruction effects, e.g., Swedish (Klingvall, 2018).

⁶In general, Thai movement dependencies respect islands. As discussed in Jenks (2014) (see footnotes 4 and 5 in Jenks 2014), this conclusion is different from what Hoonchamlong (1991) comes to. However, we believe that Hoonchamlong’s island-insensitive data may ultimately have the same explanation as what we propose for the *tough*-construction: there is a different left-peripheral position that is generated without movement.

In sum, on the one hand, the Thai *tough*-construction exhibits some recognizable properties, like thematic displacement (i.e., the defining property), nonfiniteness, and a lack of reconstruction. On the other hand, the Thai *tough*-construction appears to be exceptional in that it permit non-movement dependencies which can target potentially any position in the embedded clause.

What should we make of these facts? One option is to simply admit that the *tough*-construction in Thai is a different kind of creature from most other reported cases of *tough*-constructions—and from other dependency chains as well. Ideally, we might attribute the differences to parametric variation between languages. We would then want to know what properties Thai has that allows it to create these kinds of “exceptional” *tough*-constructions.

We agree in principle with this approach, but we will double-down on the idea that the *tough*-construction in Thai is not “exceptional”. We show below that the sentence in (13), repeated from above, is actually ambiguous.

- (13) [tôn máy tôn nủ]₁ yâak [thîi cà? tàt ____₁]
 tree CLF this difficult COMP IRR cut
 ‘This tree is difficult to cut.’

The ambiguity in (13) lies in the syntactic status of the phrase *tôn máy tôn nủ* ‘this tree.’ On the one hand, it is a “true” subject, in Spec,TP (or equivalent). On the other hand, *tôn máy tôn nủ* ‘this tree’ is a broad (or major) subject. As we show below, broad subjects in Thai are merged *in situ* and do not show any sort of movement-sensitive properties.

Why should we care about this distinction? There are two reasons. First, the observation that Thai has a broad subject has not been previously documented, and it provides a more accurate understanding of Thai syntax in general. Second, correctly analyzing the Thai *tough*-construction leads to a more accurate picture of the *tough*-construction in general. While much of the crosslinguistic discussion of *tough*-constructions has focused on the variation, in the final part of this paper, we will discuss what we believe are the universal characteristics of all *tough*-constructions. As we observe below, these core properties still require an explanation.

3 Thai Broad Subjects

In this section, we will motivate an articulated view of “subjecthood” in Thai. We start with the observation that Thai is a language that allows so-called “double-subject constructions”, where two DPs can occur in the subject position (Iwasaki, 2002; Piriya-wiboon, 2011). In these constructions, an initial DP (DP₁) can be followed by a pronoun (DP₂) that is coreferential with DP₁ or another DP.

- (14) [mamuang ton nîi]_{DP1} [luuk]_{DP2} dok dii
 mango CL.tree this fruit be.many PART
 ‘This mango tree (topic), it produces a lot of mangoes.
 (lit. ‘This mango tree, fruits are plenty.)’ (Piriya-wiboon, 2011, p.14)
- (15) Sam_{DP1} mêe_{DP2} pen nâkphaasasàat
 Sam mother BE linguist
 ‘Sam is such that his mother is a linguist.’
- (16) Sam_{DP1} khăw_{DP2} sâu rôt khan nîi
 Sam 3.SG buy car CLF this
 ‘Sam is such that he bought this car.’

While there is previous work on Thai topics (Hoonchamlong, 1991; Iwasaki and Ingkaphirom, 2005), the core—and new—idea that we will demonstrate is that Thai has a position between topic and narrow subject: the broad subject (or “subject of predication” in Cardinaletti’s 2004 terms). Importantly, we will show that Thai can have sentences which include a broad subject and a *null* resumptive pronoun, that is, a sentence like (16) without the pronoun *khǎw* ‘s/he’.⁷ Thus, in Thai a sentence with a single surface “subject”, like the original example we showed in (1b), is structural ambiguous as to where that subject is sitting; it may be a narrow subject, or it may be a broad subject. Ultimately, we argue that Thai has a three-way distinction between topics, broad subjects, and narrow/thematic subjects, outlined in (17).

(17) [Topic ... [Broad Subject ... [Narrow Subject ...]]]

Broad subjects have a number of properties which differentiate them from both thematic subjects as well as topic/focus positions. These distinctions include (i) broad subjects may be non-referential (i.e., a *wh*-word or universal quantifier); (ii) broad subjects need not be strictly left-aligned with the utterance (i.e., in a topic position); and (iii) broad subjects need not be linked to a gap. The first two properties are shared with thematic subjects, while the last property is shared with topichood. Thus, a broad subject is sometimes viewed as a mix between a topic and a narrow subject. However, we note one clear distinction: narrow subjects are *derived*, meaning that some element is promoted to occupy the narrow subject position. Broad subjects, like topics, are born *in situ*.

To begin, in (18) we show that all three positions can be filled simultaneously.⁸

(18) [*phûuchaay khon nán nà* [*rót khǎw* [*prà?tuu phaj*]]]
 man CLF that TOP car 3SG door break
 ‘As for that man, his car is such that the door broke.’

Morphologically, topics are identifiable by the appearance of a topic marker, *nà* in (18) above.⁹ There is no overt morphology that marks a broad subject or narrow subject.

Semantically, topics and broad subjects both instantiate something like an “aboutness” relationship with the clause. Aboutness is in fact rather difficult to precisely define (cf, Rawlins 2013). However, the clearest indicator of aboutness is that it constrains the availability of topics and broad subjects in various ways. If the clause cannot be construed as about the topic or broad subject, the sentence sounds infelicitous.

⁷Resumptive pronouns in Thai are largely interchangeable with null pronouns (*modulo* constraining factors like animacy and definiteness; see Hoonchamlong 1991; Iwasaki and Ingkaphirom 2005). There is an alternative to (16) which uses the “expletive” pronoun *man* instead of *khǎw*.

(i) *Sam man sǔu rót khan nǐ*
 Sam EXPL? buy car CLF this
 ‘Sam is such she that bought this car.’

Whether *man* is a true expletive or not is still, we believe, up for debate. Most sources treat *man* as another kind of resumptive pronoun, usually with pejorative connotations about the antecedent when it is animate (see Kaenmuang et al. 2025 and references therein). We note this fact because it makes similar cases in the *tough*-construction difficult to parse. It is unclear whether the sequence [DP [*man* [Adjective. ...]]] involves a resumptive pronoun in Spec,TP or an expletive subject. On our analysis, in fact, both are possible.

⁸A note on translation: since English does not have a broad subject position, we have found it difficult to accurately translate broad subjects as distinct from topics and narrow subjects. We have chosen the paraphrase “such that”, noting that this may not be an exact reflection of the semantics of a broad subject. Our translation keeps broad subjects distinct from topics (translated with “as for”) and narrow subjects (which have no unique indicators).

⁹It has been documented that Thai has five topic markers, namely *nán*, *nǐ*, *ní*, *nía* and *nà*. We refer the reader to Iwasaki and Ingkaphirom (2005, sec 30.1) and references therein for discussion of what differentiates these topic markers. Additionally, as discussed in Hoonchamlong (1991); Iwasaki and Ingkaphirom (2005), the topic marker is sometimes optional (for reasons which are not fully clear to us), in which case topichood must be diagnosed either prosodically—topics are prosodically separated while broad subjects are not—or with one of the methods outlined below.

- (19) a. # *Sam nà, Bill chàlàat*
 Sam TOP Bill intelligent
 ‘#As for Sam, Bill is intelligent’
 b. # *Sam Bill chàlàat*
 Sam Bill intelligent
 ‘#Sam is such that Bill is intelligent’

Relatedly, broad subjects and topics have a similar restriction in that they must bind some “correlate” in their scope. The correlate can be a (null) pronoun and in some cases, it can be entirely implicit (cf, Villa-García 2023).¹⁰

- (20) a. *cháaŋ ɲuaŋ yaaw*
 elephant trunk long
 ‘The elephant – its trunk is long.’ (Iwasaki and Ingkaphirom, 2005, 360)
 b. *Sam (nà) bâan (khǎw) sǎay máak*
 Sam TOP house 3SG beautiful very
 ‘As for Sam,/Sam is such that his house is very beautiful.’
 c. *ɲaan tɛŋɲaan Noy, [s Nít tòkloŋ cà? tittɔɔ rǎaŋ càt dɔɔkmáy]*
 event wedding ‘Noy’ ‘Nít’ agree ASP contact matter arrange flower
 ‘As for Noy’s wedding, Nít agrees to make arrangements for the flowers.’ (Hoonchamlong, 1991, 90)

Importantly, note that the correlate can, in principle, be anywhere within the clause, even within an island (cf, Hoonchamlong 1991). We take this to indicate that neither topics nor broad subjects involve a movement dependency in Thai.

- (21) a. *Sam₁ (nà) Bill dâyyin khàawlɛɛ [CP thîi khǎw₁ sǔu bâan mǎy]*
 Sam TOP Bill hear rumor COMP 3SG buy house new
 ‘As for Sam,/Sam is such that Bill heard the rumor that he bought a new house’
 b. *Sam₁ (nà), dekdek sǎacay phrɔ́? chǎn dǎa khǎw₁*
 Sam TOP children sad because 1SG scold 3SG
 ‘As for Sam,/Sam is such that children were sad because I scolded him’
 c. *Ben₁, c^hán dâyyin k^hà:w wâ: (k^háw₁) tɛ:ŋɲan*
 Ben I hear news COMP he marry
 ‘Ben₁, I heard the news that he₁ married.’ (Ruangjaroon, 2005, 161)

In sum, broad subjects and topics share some similarities. Importantly, broad subjects and narrow subjects also share a number of properties. In fact, if there is just one pre-verbal DP, it is not possible to distinguish whether it is a broad or narrow subject without additional diagnostics.

- (22) *Sam sǔu rɔt khan nǐ*
 Sam buy car CLF this
 ‘Sam bought this car’ or ‘Sam is such that he bought this car.’

¹⁰The presence or absence of an overt resumptive pronoun is independent of topics and broad subjects, and is sensitive to a variety of factors including animacy and definiteness (Iwasaki and Ingkaphirom, 2005). We put this complication aside here.

In terms of speaker intuitions, this sentence is ambiguous because *Sam* can be understood as a narrow subject or a broad subject. In the latter case, the intuition is that the sentence is “about” Sam in some way. Syntactically, the issue is that the sentence can be given either of the two parses below. (Note that Thai is well-documented for employing null subject pronouns in certain configurations; Hoonchamlong 1991; Iwasaki and Ingkaphirom 2005.

- (23) a. [*Sam sǔu rǒt khan nǐ*]
 Sam buy car CLF this
 ‘Sam bought this car.’ Narrow subject parse
- b. [*Sam₁ [*pro₁ sǔu rǒt khan nǐ*]]*
 Sam buy car CLF this
 ‘Sam is such that he bought this car.’ Broad subject parse

However, this distinction is subtle, and so we will provide below various empirical diagnostics we have found to differentiate between broad and narrow subjects—as well as topics. We will then turn to applying the relevant diagnostics to the *tough*-construction.

3.1 Ordering of parentheticals

We provide five diagnostics that differentiate narrow subjects from broad subjects and topics.¹¹ First, parenthetical expressions can delineate the narrow subject from the higher two positions. In (24), we show that the parenthetical expressions *nay khwaamkhít khǒng Bill* ‘in Bill’s opinion’ may occur after a topic or after a broad subject, but not after a narrow subject.

- (24) a. *phǔuchaay khon nán nà, (nay khwaamkhít khǒng Bill) rǒt khǒng khǎw prà?tuu phaj*
 man CLF that TOP in opinion of Bill car of 3SG door broke
 ‘As for that man, (in Bill’s opinion) his car is such that the door broke.’ (After topic)
- b. *phǔuchaay khon nán nà, rǒt khǒng khǎw (nay khwaamkhít khǒng Bill) prà?tuu phaj*
 man CLF that TOP car of 3SG in opinion of Bill door broke
 ‘As for that man, his car (in Bill’s opinion) is such that the door broke.’ (After broad subject)
- c. * *phǔuchaay khon nán nà, rǒt khǒng khǎw prà?tuu (nay khwaamkhít khǒng Bill) phaj*
 man CLF that TOP car of 3SG door in opinion of Bill broke
 ‘As for that man, his car is such that the door (in Bill’s opinion) broke.’ (After narrow subject)

Thus, if a parenthetical can follow a pre-verbal DP, then the DP must be a broad subject or topic.

3.2 Idioms

Idiomatic expressions can also be used to distinguish true narrow subjects from topics and broad subjects. In (25) we show that positioning a VP-internal portion of an idiom in a pre-sentential position, either as a

¹¹We note one test that is used in Cardinaletti (1990) to distinguish broad subjects from narrow subjects: so-called “weak” pronouns in Indo-European languages must be narrow subjects. However, in Thai, there is no “weak” pronominal series, and as a rule, a pronoun can be a topic or broad subject (as well as a narrow subject).

- (i) *khǎw (nà) rǒt phaj*
 3SG TOP car break
 ‘As for him/He is such that his car broke down.’

topic or broad subject, only yields a literal interpretation. This tells us that topics and broad subjects do not reconstruct for idiomatic interpretation.

- (25) a. *Sam plòy kày*
 Sam release chicken
 ‘Sam accidentally revealed the secret.’ (Lit: ‘Sam released the chicken.’) ✓Idiomatic reading
- b. *kày₁ (nà) Sam plòy ____₁*
 chicken TOP Sam release
 ‘As for the chicken/the chicken is such that Sam released it’ ✗Idiomatic reading

Some idioms include the subject as part of the idiomatic expression. In these cases, if the subject is topic-marked as in (26b) or pronoun-doubled as in (27b) the idiomatic meaning disappears. (Note that we use an animate subject in (27b) to control for certain orthogonal aspects of resumptive pronouns in Thai.)

- (26) a. *lûukmây lòn mây klay tôn*
 nut fall NEG far tree
 ‘Children are often similar to their parents (in personality or other traits).
 (Lit: ‘The nut does not fall far from the tree’) ✓Idiomatic reading
- b. *lûukmây nà lòn mây klay tôn*
 nut TOP fall NEG far tree
 ‘As for the nut, it does not fall far from the tree’ ✗Idiomatic reading
- (27) a. *kràtàay tûun tuum*
 rabbit be.startled stone-apple
 ‘The person panics easily/is falsely alarmed.’ ✓Idiomatic reading
 (Lit. The rabbit is startled by a stone apple)
- b. *kràtàay₁ man₁ tûun tuum*
 rabbit RSM be.startled stone-apple
 ‘The rabbit is such that it is startled by a stone apple’ ✗Idiomatic reading

The core takeaway is that neither topics nor broad subjects can receive an idiomatic interpretation together with the verb phrase. Only narrow subjects can be included in the idiom.

3.3 Embedded clauses

It is commonly found that subordinate clauses restrict the possibilities of the left-peripheral structure. In Thai, under the complementizer *thîi* (here introducing finite clauses), only narrow subjects are possible. (29a) demonstrates that “double subjects” are not permitted, and (29b) and (29c) show that topics are not permitted.

- (28) *Sam (nà) khǎw sǔu rót khan nǐ*
 Sam TOP 3SG buy car CLF this
 ‘As for Sam,/Sam is such that he bought this car.’
- (29) a. *chǎn chûa [CP thîi Sam (*khǎw) sǔu rót khan nǐ]*
 I believe COMP Sam 3SG buy car CLF this
 ‘I believe that Sam bought this car.’

- b. * *chǎn chûa* [_{CP} *thîi Sam na (khǎw) sǔu rôt khan nǐ*]
 I believe COMP Sam TOP 3SG buy car CLF this
 ‘I believe that Sam bought this car.’
- c. * *chǎn chûa* [_{CP} *thîi rôt khan nǐ nà, Sam cà? kháp*]
 I believe COMP car CLF this TOP Sam IRR drive
 Intended: ‘I believe that as for this car, Sam will drive it.’

Similarly, we find that sentential subjects—also headed by *thîi*—only allow a surface narrow subject, (30).

- (30) a. [_{CP} *thîi Sam (*khǎw) sǔu rôt khan nǐ*] *pen khàaw dīi*
 COMP Sam 3SG buy car CLF this COP news good
 ‘That Sam bought this car is good news.’
- b. * [_{CP} *thîi Sam nà (khǎw) sǔu rôt khan nǐ*] *pen khàaw dīi*
 COMP Sam TOP 3SG buy car CLF this COP news good
 Intended: ‘*That as for Sam, he bought this car is good news.’

As it will be important in a moment, note that Thai has another complementizer, *wâa*, which does permit higher structure, (31). (See discussions in Singhapreecha 2010 and Jenks 2011, 2014.)

- (31) a. *chǎn chûa* [_{CP} *wâa Sam khǎw sǔu rôt khan nǐ*]
 I believe COMP Sam 3SG buy car CLF this
 ‘I believe that Sam is such that he bought this car.’
- b. *chǎn chûa* [_{CP} *wâa rôt khan nǐ nà, Sam cà? kháp*]
 I believe COMP car CLF this TOP Sam IRR drive
 ‘I believe that as for this car, Sam will drive it.’

The two complementizers differ not only in their syntax but also in their semantics (den Dikken and Singhapreecha, 2004; Jenks, 2014). Broadly, *thîi*-clauses correlate with nominal syntax—*thîi* is used in most adnominal “linking” dependencies. *Thîi*-clauses are associated with factive readings and sentential subjects must be headed by *thîi*. In contrast, *wâa* is associated with verbal syntax—it is independently a verb meaning ‘say.’ In contrast to *thîi*-clauses, *wâa*-clauses are found in nonfactive contexts. We note finally that the complementizers may co-occur in the order *thîi wâa*. These differences will not be critical in the data and analysis below.

3.4 Long Distance Anaphors

The fourth factor that distinguishes narrow subjects from the two higher positions concerns the distribution of long distance anaphors. In Thai, the long distance anaphor *tua?eeŋ* can be the subject in an embedded clause, as shown in (32). (See discussion of Thai anaphors in Meechanyakul 2024.)

- (32) *Sam chûa* [_{CP} *wâa/thîi tua?eeŋ cà? cháná? kaanlǎaktâŋ*]
 Sam believe COMP REFL FUT win election
 ‘Sam₁ believes that SELF₁ will win the election.’

As we demonstrate in (33), these subject long distance anaphors must be narrow subjects; a long distance anaphor cannot be bound when placed somewhere higher than the lowest subject position. (Here we use a *wâa* clause to allow for the possibility of topic and broad subject positions.)

- (33) a. **Sam phũut* [_{CP} *wâa tuaʔeeŋ mǎa pùay*]
 Sam say COMP REFL dog sick
 ‘Sam₁ said that SELF₁ is such that his dog is sick’
 b. **Sam phũut* [_{CP} *wâa tuaʔeeŋ nà, mǎa pùay*]
 Sam say COMP REFL TOP dog sick
 ‘Sam₁ said that as for SELF₁, his dog is sick’

3.5 Non-referential subjects

Finally, a narrow subject can be entirely non-referential, i.e., an expletive. Neither broad subjects nor topics can be expletives. This is easy to show for topics in (34, 35), because an expletive cannot be topic-marked.

- (34) a. *man rón*
 EXPL hot
 ‘It’s hot’
 b. **man (nà) ʔaakàat rón*
 EXPL TOP weather hot
 ‘*As for it/It is such that the weather is hot’
 (35) a. *man mũut*
 EXPL dark
 ‘It’s dark’
 b. **man (nà) klaaŋkhũn mũut*
 EXPL TOP night dark
 ‘*As for it/It is such that the night is dark.’

The ban on expletive broad subjects is more difficult to show, but it is primarily motivated on semantic grounds. Broad subjects require an aboutness relationship between the subject and the proposition. A proposition cannot be about a meaningless element.

3.6 Quantificational subjects

We include one (well-known) factor that distinguishes topics from broad and narrow subjects: a broad or narrow subject can be quantificational, but a topic cannot. Unlike Thai (and English) topics, a pre-sentential element can be *wh*-questioned and can be universally quantified—but neither can be topic-marked, (36).

- (36) a. *kray (*nà) rôt hǎay mũawaan*
 who TOP car disappear yesterday?
 ‘Who is such that their car disappeared yesterday?’ (Not ‘*As for who, their car disappeared yesterday?’)
 b. *nákrian thúk khon (*nà) rôt hǎay mũawaan*
 student every class TOP car disappear yesterday
 ‘Every student is such that their car disappeared yesterday.’ (Not: ‘*As for every student, their car disappeared yesterday.’)

	Overt marker	Paren. following	Idiom	In <i>thîi</i> clause	Subject Anaphor	Nonref. Subj	Quantifier
Topic	✓	✓	*	*	*	*	*
Broad subject	*	✓	*	*	*	*	✓
Narrow subject	*	*	✓	✓	✓	✓	✓

Table 1: Summary of distinctions between topics, broad subjects, and narrow subjects in Thai

3.7 Summarizing

We summarize the various distinctions between topics, broad subjects, and narrow subjects in Table 1.

We conclude from this data that Thai has a broad subject position, which is hierarchically situated between a topic and narrow subject position.

(37) [Topic ... [Broad Subject ... [Narrow Subject ... [...]]]

Importantly, a sentence can have a broad subject without an overt narrow subject. That is, (38a) is syntactically—as well as semantically—ambiguous. This is evident from the fact that a parenthetical can follow a single preverbal DP, (38b). (Recall that parentheticals cannot follow narrow subjects.)

- (38) a. *Sam sǔu rôt khan nǐi*
 Sam buy car CLF this
 ‘Sam bought this car’ or ‘Sam is such that he bought this car.’
- b. *Sam, nay khwaamkhít khǎw Bill, sǔu rôt khan nǐi*
 Sam in opinion of Bill buy car CLF this
 ‘Sam is such that, in Bill’s opinion, he bought this car.’

We assume that in the cases where there is just a broad subject and no overt narrow subject, there is a null resumptive pronoun in the subject position.

4 Differentiating broad subjects and narrow subjects in the *tough*-construction

In the next few sections, we will apply some of the diagnostics that uniquely pick out a narrow subject in Thai to the original sentence in (1b), showing that the *tough*-construction in Thai does in fact involve a movement dependency. We note that not all diagnostics apply in the *tough*-construction, for reasons we will briefly discuss in subsection 4.4.

4.1 Two subjects

While a sentence with a single non-topic marked DP in front of the predicate is ambiguous with respect to whether it is a narrow subject or a broad subject, a sentence with two preverbal non-topic DPs is not;

the second DP must be a narrow subject. As shown in (39), in the *tough*-construction, when both the broad subject and the narrow subject positions are overtly filled, island sensitivity of the *tough*-construction resurfaces, restricting the second DP—the narrow subject.¹²

(39) *Adjunct island*

- a. [rót khan nĩ]₁ yâak mâak thĩ Jane cà? sacay [CP phrɔʔ Sam sũũ ____₁]
 car CLF this difficult very COMP Jane IRR upset because Sam buy
 ‘This car₁ is such that it₁ is very difficult for Jane to be upset because Sam bought it.’
- b. [nákrian khon nĩ]₁ man yâak mâak thĩ Jack cà? sacay [CP phrɔʔ Sam sũũ rót
 student CLF this EXPL difficult very COMP Jack IRR upset because Sam buy car
 khăw₁]
 her
 ‘This student is such that it is very difficult for Jack to be upset because Sam bought her car.’
- c. * [nákrian khon nĩ]₁ [rót (khăw₁)]₂ yâak mâak thĩ Jack cà? sacay [CP phrɔʔ Sam
 student CLF this car 3SG difficult very COMP Jack IRR upset because Sam
 sũũ ____₂]]]
 buy
 ‘*This student is such that her car is very difficult for Jack to be upset because Sam bought ____.’¹³

(40) *Sentential subject*

- a. [rót khan nĩ]₁ yâak mâak thĩ cà? khít wâa [CP thĩ Sam cà? khăy ____₁] pen
 car CLF this difficult very COMP IRR think COMP COMP Sam IRR sell COP
 khàaw dii
 news good
 ‘This car is such that it is very difficult to think that that Sam will sell it is good news.’
- b. * [nákrian khon nĩ]₁ [rót (khăw₁)]₂ yâak mâak thĩ cà? khít wâa [CP thĩ Sam
 student CLF this car 3SG difficult very COMP IRR think COMP COMP
 cà? khăy ____₂] pen khàaw dii
 Sam IRR sell COP news good
 ‘*This student is such that her car is very difficult to think that that Sam will sell is good news.’

In prose, while a single initial DP can be a broad subject and therefore is free bind any position within its scope, the second of two initial DPs is syntactically constrained. The leftmost DP occupies the broad subject position, and so the following DP must be a narrow, i.e., derived subject. When this subject is coindexed with a gap in an island, the sentence is ungrammatical.

We show the same idea with subject gaps as well. In (41), we show that a single pre-verbal DP can target a subject position in the embedded clause. But when two DPs are present, then subject gaps are

¹²We are using the convention of the underline “____” to indicate “silence”, abstracting away from whether that silence is due to movement or whether it is a silent pronoun *pro*.

¹³The *tough*-construction without the island violation is grammatical as shown below.

(i) [nákrian khon nĩ]₁ [rót (khăw₁)]₂ yâak mâak thĩ cà? sũũ ____₂ nay raakhaa \$200
 student CLF this car 3SG difficult very COMP IRR buy in price \$200
 ‘This student is such that her car is very difficult to buy ____ for \$200.’

ungrammatical in the *tough*-construction, consistent with English. The first of two subjects—the broad subject—can target a subject position, but the second—the narrow subject—cannot.¹⁴

- (41) a. *dekdek₁ man yâak [thîi ____₁ cà? khâwcay phaasăa-angkrit]*
 children EXPL difficult COMP IRR understand language-English
 ‘Children are such it is difficult for them to understand English.’
 b. *dekdek₁ phaasăa-angkrit₂ yâak [thîi ____₁ cà? khâwcay ____₂]*
 children language-English difficult COMP IRR understand
 ‘Children are such that English is difficult for them to understand ____.’
 c. **phaasăa-angkrit₂ dekdek₁ yâak [thîi ____₁ cà? khâwcay ____₂]*
 language-English children difficult COMP IRR understand
 ‘*English is such that children are difficult ____ to understand it.’

We note that reversing the order of the binding relationship is grammatical. As we show in (42a). When the leftmost of two pre-verbal DPs binds the subject gap position in the embedded clause, the sentence is fine. (In (42a), Sam is most naturally understood as the one who sells the car.) But simply switching the order of these DPs is ungrammatical (42b) because now the narrow subject binds the subject position in the *tough*-construction.

- (42) a. *Sam₁ rôt₂ yâak thîi ____₁ cà? khăay ____₂*
 Sam car difficult COMP IRR sell
 ‘Sam is such that his car is difficult for him to sell ____.’
 b. **[rôt khan nîi]₂ Sam₁ yâak thîi ____₁ cà? khăay ____₁*
 car CLF this Sam difficult COMP IRR sell
 ‘*This car is such that Sam is difficult ____ to sell it.’

As these examples show, the problem is not simply having two subjects in a *tough*-construction. Rather, the issue is the correlation between the position of the subject and the gap.

4.2 Embedded clauses

Recall that under the complementizer *thîi*, only a narrow subject is permitted. When a *tough*-construction is embedded in a *thîi*-clause, again we find that the island sensitivity returns. This can be compared to embedding under a *wăa*-clause, (43b), which as expected allows island insensitive *tough*-constructions because *wăa* clauses allow embedded broad subjects (and topics).

- (43) a. **chăn chûa [CP thîi [năŋsău lêm nîi]₁ yâak [thîi cà? pay hōŋsà?mùt phûa*
 1SG believe COMP book CLF this difficult COMP IRR go library for
thîi cà? pay ?àan ____₁]
 COMP IRR go read
 ‘*I believe that this book is difficult to go to the library in order to read ____’

¹⁴A reviewer notes that this effect could be due to the fact that (41c) has a nested dependency. However, nested dependencies are not otherwise banned in Thai, though it is possible that there is a more subtle constraint which takes into consideration the kind of dependencies that are nested.

- b. *chăn châu* [_{CP} *wâa* [*năŋsũu lêm nũ*]₁ *yâak* [*thũi* *cà? pay hōŋsà?mùt phũa*
 1SG believe COMP book CLF this difficult COMP IRR go library for
thũi *cà? pay ?àan* ____₁]]
 COMP IRR go read
 ‘I believe that this book is such that it is difficult to go to the library in order to read it.’

In addition, the *thũi*-clause embedded under *châu* ‘believe’ fails to embed a *tough*-construction that has both DP subject (broad subject) and the expletive subject (narrow subject) as in (44).

- (44) **Sam châu* [_{CP} *thũi* *dèkdèk*₁ *man yâak thũi* *cà? sǝn* ____₁]
 Sam believe COMP children EXPL difficult COMP IRR teach
 ‘Sam believes that children are such that it is difficult to teach them.’
- (45) *Sam châu* [_{CP} *wâa* *dèkdèk*₁ *man yâak thũi* *cà? sǝn* ____₁]
 Sam believe COMP children EXPL difficult COMP IRR teach
 ‘Sam believes that children are such that it is difficult to teach them’

A similar example is shown in (46), where we have a subject CP. Inside of the subject CP, island effects hold for the subjects of the *tough*-construction.

- (46) a. [_{CP} *thũi* [*năŋsũu lêm nũ*]₁ *yâak thũi* *cà? ?àan* ____₁ *nay hōŋsà?mùt*] *pen ruângkoohòk*
 COMP book CLF this difficult COMP IRR read in library COP lie
 ‘That this book is difficult to read ____ in the library is a lie’.
- b. * [_{CP} *thũi* [*năŋsũu lêm nũ*]₁ *yâak* [_{CP} *thũi* *cà? pay hōŋsà?mùt* [_{CP} *phũa thũi* *cà?*
 COMP book CLF this difficult COMP IRR go library for COMP IRR
pay ?àan ____₁]]] *pen ruângkoohòk*
 go read COP lie
 ‘*That this book is difficult to go to the library in order to read ____ is a lie.’

Likewise, the ban on subject gaps is again enforced under *thũi*, whether as complement clause or subject clause. *Wâa* clauses (which cannot be subjects), on the other hand, allow apparent *tough*-constructions targeting subject positions.

- (47) a. **chăn châu* [_{CP} *thũi* *dekdek*₁ *yâak* [*thũi* ____₁ *cà? khâwcay phaasǝa-angkrìt*]]
 1SG believe COMP children difficult COMP IRR understand language-English
 ‘*I believe that children are difficult ____ to understand English.’
- b. * [_{CP} *thũi* *dekdek*₁ *yâak* [*thũi* ____₁ *cà? khâwcay phaasǝa-angkrìt*]] *pen*
 COMP children difficult COMP IRR understand language-English COP
ruângkoohòk
 lie
 ‘*That children are difficult ____ to understand English is a lie.’
- c. *chăn châu* [_{CP} *wâa* *dekdek*₁ *yâak* [*thũi* ____₁ *cà? khâwcay phaasǝa-angkrìt*]]
 1SG believe COMP children difficult COMP IRR understand language-English
 ‘*I believe that children are such that it is difficult for them to understand English.’

4.3 Long Distance Anaphors

Recall that subject long-distance anaphors must be narrow subjects. We therefore expect that when a long-distance anaphor is the subject of a *tough*-construction, it will be sensitive to movement constraints and cannot bind a subject gap. As we can see in (48a) and (48b), this is correct. (Note that we are using the complementizer *wâa*, which can embed higher functional structure, so the restriction is not due to the clause size.)

- (48) *Sam₁ phûut [CP wâa tuaʔeeŋ₁ yâak thîi phôw cà khâwcaŋ ____₁*
 Sam say COMP REFL difficult COMP father IRR understand
 ‘Sam₁ said that SELF₁ is difficult for his father to understand ____.’
- a. **Sam₁ phûut [CP wâa tuaʔeeŋ₁ yâak thîi càʔ dâyyin [NP khâawlum [CP thîi phôw*
 Sam say COMP REFL difficult COMP IRR hear rumor COMP father
mây khâwcaŋ ____₁]]
 NEG understand
 Intended: ‘*Sam₁ said that SELF₁ is difficult to hear a rumor that his father does not understand ____.’
- b. **Sam₁ phûut [CP wâa tuaʔeeŋ₁ yâak thîi ____₁ càʔ khâwcaŋ phaasâa-angkrit*
 Sam say COMP REFL difficult COMP IRR understand language-English
 Intended: ‘*Sam₁ said that SELF₁ is difficult ____ to understand English.’

4.4 Summarizing

Summarizing what we have discovered above, Thai has two subject positions, a narrow subject and a broad subject. This distinction can be obscured in certain contexts, leading to the false impression that Thai’s *tough*-construction exceptionally allows unconstrained long-distance dependencies and subject gaps. However, we have shown above that when we control for the various subject positions and force the subject to be a narrow subject, then the *tough*-construction in Thai more closely resembles the crosslinguistic picture in that there is a movement dependency with the gap originating in a non-subject position.

It is important to note that not all diagnostics that distinguish broad and narrow subjects in Thai can be applied in the *tough*-construction to distinguish between subject positions. For instance, we observed that expletives must be narrow subjects. Unfortunately, expletives cannot appear as the subject in the *tough*-construction in general; this is an independent constraint, active even in English (Lasnik and Fiengo, 1974). We also observed that idioms in which the subject is part of the idiom require that the subject be a narrow subject. Again, this point is moot in the *tough*-construction since the subject of the *tough*-predicate does not form an idiomatic expression with the *tough*-predicate. And finally, while parentheticals can be used to identify broad subjects when there are multiple pre-verbal DPs, this diagnostic is not useful in determining subjecthood in the *tough*-construction because putting a parenthetical *after* a narrow subject will be ungrammatical, and so any additional tests (like islandhood) are moot. On the other hand, putting a parenthetical *before* a single pre-verbal DP is compatible with both broad and narrow subject parses.

Before turning to the analysis, we will mention two other languages that have been reported to have similar unconstrained antecedent-gap dependencies in the *tough*-construction. Crucially, both languages have well-documented broad subject positions, as well. Tayalati and Danckaert (2020) argue that this is the way to understand so-called “resumptive” *tough*-constructions in Modern Standard Arabic, (49a) and (49b).¹⁵

¹⁵A third language that has argued to be “mis-analyzed” in the same way is Modern Greek. Alexiadou and Anagnostopoulou (2020)

- (49) a. *hā'ulā'i n-nisā'-u darūriyy-un 'iqnā'u=hunna*
 those.F.PL the-woman(F.PL-NOM necessary.M.SG-NOM convince(M).SG-NOM=RP.F.PL
 'Those two women are necessary to convince.'
 b. *hādāni l-kitāb=āni₁ mumti'-un qirā'at-u=humā₁*
 those.M.DU the-book(M).DU-NOM pleasant.M.SG-NOM read(F).SG-NOM=RP.M.DU
 'Those two books are pleasant to read.'

(Tayalati and Danckaert, 2020, 201)

According to Tayalati and Danckaert, the subjects in (49) are broad subjects binding an obligatory resumptive pronoun on the embedded deverbal nominalization. There is a silent expletive *pro* in Spec,TP, and it is this lower implicit subject that triggers obligatory masculine singular agreement on the adjective. This analysis is independently supported by the fact that Modern Standard Arabic is well-documented as having a broad subject position (Doron and Heycock, 1999; Heycock and Doron, 2003; Alexopoulou et al., 2004). Strikingly then, Modern Standard Arabic, unlike Thai, does not have a “true” *tough*-construction. Rather it has a broad subject construction, which superficially resembles a *tough*-construction.

A more complicated (and controversial) set of data has been widely discussed in Japanese (Inoue, 1978, 2004; Ohkado, 1993; Takita et al., 2022). In general, Japanese *tough*-constructions are well-known to be island-insensitive, part of a larger pattern of island-violating dependencies in Japanese (Kuno, 1973; Saito, 1985; Takezawa, 1987).¹⁶

- (50) *konote-no hanzai-ga₁ keisatu-nitotte [vP [____₁ okasita] ningen]-o*
 this.kind-GEN crime-NOM keisatu-nitotte committed man-ACC
sagasi]-niku-yasu-i
 search.for-tough/easy-is
 ≈ This kind of crime₁ is tough/easy for the police to search for a man who committed ____₁ Takita et al. (2022, 260) adapted from Takezawa (1987)

Importantly, just like in Thai, when we preclude the possibility of a broad subject position for the subject of the *tough*-construction, island-sensitivity returns. In (51c), we show that when we give the *tough*-construction two subjects, the lower subject respect islands.¹⁷

- (51) a. *[Kono tosyokan-kara]-ga₁ [____₁ hon-o nusumi]-niku/yasu-i*
 This library-from -NOM book-ACC steal -tough/easy-is
 (lit) 'From this library is tough/easy to steal books.'

propose that the rather peculiar properties (as compared to English) of Greek *tough*-constructions make sense if the “subject” is actually a kind of topic-argument, binding a null pronoun.

¹⁶This is a slight over-simplification since post-positional subjects do exhibit island effects in some cases. We note that Takita et al. (2022) distinguish between “*vP*-tough” and “*CP*-tough”. The former has a smaller complement clause than the latter.

¹⁷Thanks to Nicholas Kamyu Mori for the Japanese data in (51). Additionally, island effects re-appear when the subject is forced to reconstruct lower than its surface position.

- (i) * *Zibun-no₂ runbun-ga₁ kyoosi-nitotte₂ [vP [zyugyoo-de ____₁ hihansita] gakusei]-o hyookasi]-niku/yasu-i*
 self-GEN paper-NOM teacher-for class-in criticized student-ACC evaluate-tough/easy-is

Intended: 'Self's paper is tough for teachers to evaluate a student who criticized in class.' (Takita et al., 2022, 262)

This follows because narrow subjects are derived subjects, moving to the subject position and hence can in principle reconstruct. Broad subjects are generated *in situ* and have no lower position to reconstruct to. We note that an anonymous reviewer believes that the five diagnostics we have offered for Thai work in Japanese as well to distinguish “true” *tough*-constructions from broad subject and topic constructions, though we will leave this for future work.

- b. [*Kono tosyokan-kara*]-ga₁ *hon-ga*₂ [____₁ ____₂ *nusumi*]-niku/yasu-i
 This library-from -NOM book-NOM steal -tough/easy-is
 (lit) ‘From this library books are tough/easy to steal ____.’
- c. * [*Kono tosyokan-kara*]-ga₁ *hon-ga*₂ [*Taro-ga* ____₁ ____₂ *nusumi to-iu uwasa-o*
 this library-from NOM book-NOM Taro-NOM steal COMP rumor-ACC
kiki]-niku/yasu-i
 hear -tough/easy-is
 ‘*From this library books are easy to hear the rumor that Taro steals.’

Presumably, (51c) is ungrammatical for the same reason as in Thai: the second subject is forced to be a narrow subject, and cannot freely bind a pronominal correlate in the embedded clause.

5 Analysis

Our analysis of the *tough*-construction proper (i.e., the narrow subject version) will adopt what has already been proposed elsewhere. Specifically, we assume a predication analysis of the *tough*-construction, meaning that the null-operator movement in the nonfinite clause creates a predicate which is predicated of the subject in the main clause (Browning, 1987; Heycock, 1994; Keine and Poole, 2017; Gluckman, 2021). In (52), the subject is generated in the main clause as a non-thematic argument of the verb *yâak* ‘be difficult.’¹⁸ The nonfinite clause, as a property of individuals, combines directly with the adjective, also a property of individuals.

- (52) [TP *rôt khan nîi* [VP ____ *yâak* [CP *Op*₁ *thîi* *cà? khăay* ____₁]]]
 car CLF this difficult COMP IRR sell
 ‘This car is difficult to sell ____.’

We opt for a predication analysis over a movement analysis (where the subject moves out of the nonfinite clause; Hicks 2003) since there are no reconstruction effects in Thai *tough*-constructions—but nothing crucial hinges on this.¹⁹ What is important, however, is that in this derivation there is a movement step in the nonfinite clause and the narrow subject, in Spec,TP, is the head of this chain. The subject needs the nonfinite clause to provide a thematic role, and the nonfinite clause needs the subject to bind the variable that is created via movement of the operator.

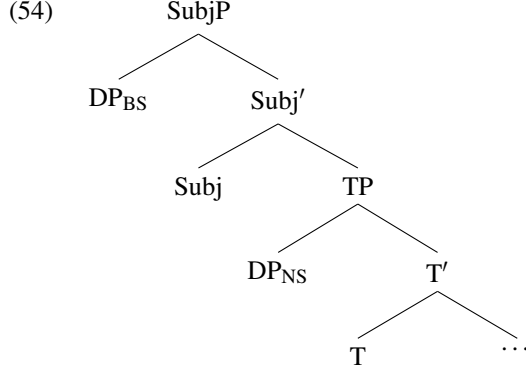
Our analysis is independently motivated by the fact that the complementizer *thîi* is also the relative complementizer. It is convincingly argued in Jenks (2014) that clauses headed by *thîi* are properties, inherently encoded to invoke operator movement. For instance, when *thîi* heads a relative clause, the relative clause is a property of individuals, which is a result of *thîi*’s lexical semantics.

- (53) [NP *nákrian*₁ [RC *thîi khruu tii* ____₁ (*mawaannîi*)]] *son mâak*.
 student COMP teacher hit yesterday naughty very
 ‘The student that the teacher hit (yesterday) is very naughty.’ (Jenks, 2014, 299)

¹⁸Note that we treat *yâak* ‘be difficult’ as a stative verb, rather than an adjective like in English. As pointed out by a reviewer, negation is a good diagnostic for the presence of a null predicate in Thai: negated copulas must be overt. But since *yâak* can be negated directly, it is more likely that it is a verbal predicate in this case.

¹⁹Interestingly, a predication-style *tough*-analysis is also employed in Sudmuk’s (2003) analysis of the *thuuk* ‘passive’ in Thai. Indeed, the constructions share many relevant properties including a long-distance dependency constrained by movement and an A’-gap headed by a subject in an A-position. In other words, the predication analysis is independently motivated in the language.

We assume following Cardinaletti (1997, 2004) *et seq* that broad subjects are introduced in a Subject Phrase (SubjP), as in (54). (It is argued to be below FinP in Cardinaletti 2004. Note that the availability of a broad subject parametrically variable across languages.)



There are two ways to understand what the head Subj does. For Cardinaletti, Subj introduces a predication relation, such that the entire clause is predicated of the broad subject. The predication relation is meant to explain the aboutness relationship between the clause and the subject. (See also related ideas in Kuno 1973; Saito 1985.) Applied to the examples of the *tough*-construction in this paper, the resulting structure is the following.

- (55) $[_{SubjP} d\grave{e}k d\grave{e}k_1 [_{TP} (man) [_{VP} y\acute{a}ak [_{CP} th\acute{i}i \quad c\grave{a}? s\check{o}n \quad pr o_1]]]]$
 children EXPL difficult COMP IRR teach
 ‘Children are such that it is difficult to teach them.’

This idea will not appeal to all syntacticians (or semanticists), because we need to constrain the Subj head from overgenerating. Specifically, we need to define what “aboutness” means so that sentences with a broad subject will generally appear with a correlate—usually a bound pronoun. There are in fact viable applications of “aboutness” that may be utilized here to sufficiently constrain this dependency; see Portner and Yabushita (1998); Rawlins (2013); Gluckman (2024) for ways to accomplish this.

An alternative more restrictive theory for Subj is that it comes with unselective binding built in: it introduces an operator that binds a variable in its scope. In this case, a dependency is formed that is not syntactically constrained—there is no movement, just variable binding. (Note this is what is also proposed to characterize left-dislocated topics in Landau 2011.)

- (56)
-
- $[_{SubjP} r\acute{o}t \text{ khan } n\acute{u}i [_{Op_1} [_{TP} (man) [_{VP} y\acute{a}ak [_{CP} th\acute{i}i \quad c\grave{a}? kh\check{a}y \quad pr^1 o_1]]]]]]$
 car CLF this 3SG.INAN difficult COMP IRR sell
 ‘This car is such that it is difficult to sell it.’

This idea works as long as we can explain away the cases in which there appears to be vacuous binding. Examples like (57) need an additional stipulation in the unselective binding approach, because there is no overt pronoun that is bound by the broad subject. There are two options that have been proposed to save the pronominal binding approach. First, Hoonchamlong (1991) argues that in all A'-dependencies in Thai in which the gap is after a preposition, the preposition is obligatorily dropped. We may make a similar claim

for the correlate pronoun in broad subject constructions, positing a null pronoun after a silent preposition in (57). That is, in (57) there is a silent “in it” in the embedded clause. (See specifically the analysis of so-called “gapless relatives” in Thai in Hoonchamlong 1991, 184ff and Jenks 2014, 310.)

- (57) [SubjP [ŋaan tɛŋŋaan Noy]₁ [TP Nit tòkloŋ cà? tittòɔ r̥uɑŋ càt dɔ̀ɔkmáy Ø pro₁]]
 event wedding Noy Nit agree ASP contact matter arrange flower in
 ‘As for Noy’s wedding, Nit agrees to make arrangements for the flowers (in it).’ adapted from
 (Hoonchamlong, 1991, 90)

Hoonchomlong appeals to the idea of a “proPP” to explain these kinds of dependencies. This is a viable idea as long as this kind of binding dependency is analyzed without movement. As we demonstrated above, broad subjects are island insensitive.

A second, related option, explored in Heycock and Doron (2003), is that the operator can bind a *situation* pronoun. On this approach, we can posit a silent situation pronoun in (57) (typically thought to be around VP) that the topic binds. This is an appealing solution as long as all of the broad subjects are plausibly construed as spatio-temporal “slices” corresponding to situations.

Both pronominal options seem viable to us, but we also believe that the “aboutness” predication relationship of Cardinaletti (2004) is also still a plausible choice—it is well motivated in some languages, at least. We do not currently see a way to tease apart these differences. (In this, we follow Hoonchamlong 1991, who similarly finds no way to differentiate between these options.) For our study, the difference is irrelevant. What is crucial is that on either analysis of broad subjects, there is no movement dependency between the broad subject and a position in its scope.

To explain restrictions on broad subjects and topics in embedded clauses, there are two options as well. A simple explanation is that *thîi*-clauses are truncated and do not project broad subject or topic positions. However, as a reviewer points out, this leads to some unanswered questions about the cartography of the left periphery as well as language-internal variation. (For instance, relative clauses headed by *thîi* do allow topics and broad subjects inside of them.) We will therefore sketch a slightly more complex solution. The core idea, building on Jenks (2011, 2014), is that *thîi* always requires something to move into its specifier. In the cases where there is no apparent gap, the grammar inserts what we call a “broad-subject operator” (Op_{BS}) in Spec,SubjP which subsequently satisfies the requirement imposed by *thîi*. We may then say that Spec,SubjP is invariably projected in *thîi*-clauses, but its specifier cannot host an overt broad subject when Op_{BS} is inserted.

This proposal explains the facts about tough-constructions presented so far. First, the broad subject can optionally be overt in the non-finite embedded clause. Because an operator associated with the object gap is already available to satisfy the requirement imposed by *thîi*, the grammar need not insert Op_{BS}, leaving Spec,SubjP available for an overt broad subject. Second, island sensitivity emerges when a tough-construction is embedded in a *thîi*-clause, as illustrated in Section 4.2. In this configuration, Spec,SubjP is occupied by the null Op_{BS}. The only overt DP available to be involved in the antecedent-gap dependency is therefore the narrow subject in Spec,TP. And finally, we are able to account for a wide range of *thîi*’s behavior, even outside of *tough*-constructions.

In closing the analysis section, we want to highlight a couple of important implications. The main-point—that the *tough*-construction is only possible with a narrow subject—tells us something about the *tough*-construction in general. In particular, the subject of the *tough*-construction is derived by A-movement *within the main clause*. This is what it means to be a narrow subject. Theories which advocate for an *in situ* analysis (e.g., Rezac 2006) are not viable because they do not distinguish between the different kinds of subject-like possibilities in a language like Thai, Arabic, and Japanese. As a reviewer correctly points out, the observation that the *tough*-construction is only possible with a derived subject does not settle the

movement vs. predication debate. There are analyses of the *tough*-construction which involve movement out of the embedded clause followed by an A-movement step in the main clause (e.g., Hicks 2003), and there are analyses which generate the subject as a true argument of the *tough*-predicate undergoing A-movement to the subject position (e.g., Gluckman 2021). Both are compatible with the observation that the subject must move to Spec,TP from a clause-internal position.

There is a second, larger implication concerning the question of *why* the *tough*-construction is restricted to derived subjects. As is well-known, *tough*-predicates like *difficult*, *easy*, *tough*, ... on their own cannot thematically license an individual-denoting subject—in English or in Thai. However, we believe that the *tough*-construction is fundamentally an example of *complex predicate formation*. The combination of the *tough*-predicate with the clause creates a new predicate, one which needs an individual-denoting subject. Again, the formal details of how this complex predicate is created may vary depending on approach (and even on language, as we discuss below). At the very least, it seems likely that this is an inherent, selectional property of *tough*-predicates: they select for particular kinds of nonfinite clauses, ones with gaps formed by movement (Keine and Poole, 2017). The end result will always be something which requires a subject to satisfy its selectional requirements, and that is why the *tough*-construction is only possible with a narrow subject. If there is no narrow subject, the selectional properties of this *ad hoc* complex predicate are not met.

6 Discussion: the *Tough*-Signature

In recent work, there has been discussion about cross-linguistic variation in *tough*-constructions (Havu and Höglund, 2015; Tayalati et al., 2020) (see also Comrie and Matthews 1990; Comrie 1997). From these works, a few differences have been identified among the languages that have a *tough*-construction. These areas of variation are (i) the size and morphological properties of the embedded clause, (ii) whether there is A- or A'-movement, (iii) the productivity of the *tough*-construction, in that some languages only have a few *tough*-predicates.²⁰ Given that there is variation among *tough*-constructions, we would now like to address what the *commonalities* are. We have made the claim above that Thai's *tough*-construction is not “exceptional”, and it fits the signature of other *tough*-constructions—but what is that signature? What are the universals, if any, that are found in all *tough*-constructions? We believe there are four core properties that all *tough*-constructions share. These are universal across all languages that have a *tough*-construction. On their own, the properties in (58) are largely unremarkable, but together conspire to create the *tough*-construction.

- (58) *The tough signature*
- a. Thematic displacement
 - b. Nonfinite embedded clause
 - c. Movement
 - d. Non-subject gap

Foremost, there is a thematic displacement. The subject in a *tough*-construction is not thematically licensed by the *tough*-predicate. This is the observation that, absent the nonfinite clause, the subject in the

²⁰To that point, it is in principle possible that a language could have a *tough*-construction without any *tough*-predicates. We believe this is what German's “Modal Infinitive” construction is teaching us.

(i) *Die Aufgaben sind zu lösen*
 the problems are to solve.INF
 ‘The problems must be solved.’

(Holl, 2010, 8)

As is widely documented, sentences like (i) share many commonalities with *tough*-constructions. While German does have a *tough*-construction proper, we may eventually find a language that *only* has the equivalent of (i).

tough-construction is not thematically licensed. This can be compared with nonfinite relative clauses (59) and the related *pretty*-construction.

- (59) a. The car was easy ??(to sell ____).
 b. The car (to sell ____) is in the garage.
 c. The car was pretty (to look at ____).

This is probably the defining property of the *tough*-construction; it is certainly the most pressing from a theoretical perspective. Empirically, it translates into the fact that there is always an alternation between a (possibly null) expletive subject and a *tough*-construction.

Second, the *tough*-construction is only possible with a nonfinite subordinate clause (Berman and Szamosi, 1972; Lasnik and Fiengo, 1974; Gluckman, 2021). Thus, strictly ruled out are cases of *tough*-constructions where the highest embedded verb is finite.²¹

- (60) a. It was important that Sam sell this car.
 b. *This car was important that Sam sell ____.

The restriction to nonfinite clauses is truly universal. There are no known counterexamples. Importantly, there is variation in the “kind” of nonfinite clause and the overall size of the clause. In Romanian, for instance, the supine form of the verb is needed (Giurgea and Soare, 2020). In Romance, it is sometimes claimed that the nonfinite verb is a “gerundive” form (Authier and Reed, 2008). Furthermore, it is clear from the Thai data that the embedded clause is a full CP, demonstrated in part by the fact that it can contain an overt subject position—clearly a related point of variation among *tough*-constructions. In contrast, Romance nonfinite clauses are smaller than CPs and cannot have overt subjects. That is, languages can vary the kind of clause used in the *tough*-construction, but the verb is *always* nonfinite.

Third, there is a movement-constrained dependency in the *tough*-construction. We find no cases in which the subject of the *tough*-construction can enter into, say, a long distance binding configuration. In all cases, *something* has moved. Here again we find crosslinguistic variation in both what has moved, and the kind of movement invoked. In Thai and English, it is quite clear that there is A'-movement (cf, Chomsky 1977 and much subsequent work for English), and it can be debated about whether this constitutes an instance of long-distance “improper movement” (e.g., Hicks 2009) or operator movement (e.g., Browning 1987). In other languages, notably Romance, there is evidence that the movement is an A-movement, similar to what is found in restructuring environments with clitic-climbing (Cardona and Villalba, 2025). What we do not find, however, are languages in which the *tough*-subject—a narrow subject—can freely bind a gap/pronoun anywhere in the embedded clause, e.g., **This author₁ is difficult for us to read her₁ books*.

Finally, the gap in the *tough*-construction is always in a non-subject position. This is of course exemplified by the original well-cited distinction between *easy* and *eager*. The former only permits the subject of *easy* to correspond to the object of the embedded verb, while the latter only displays subject control.²²

²¹In English, there is speaker disagreement about whether a finite verb is *ever* allowed to be crossed in the *tough*-construction (Dalrymple and King, 2000; Gluckman, 2021, 106).

²²Indeed, Stowell (1991) suggests that the *tough*-construction (in English) does not permit a subject gap *anywhere*. This hypothesis is difficult to test, however. Since extraction out of finite clauses is already degraded, the ungrammaticality of (i) may not be the result of the subject gap.

(i) *Benji is difficult to think (that) ____ likes anchovies.

Likewise, extraction after a *for* complementizer is also not possible for independent reasons. And finally, ECMed/Raised objects are not possible to move, but the reason for this is not clear, and again might not be related to the fact that they (possibly) involve a subject gap.

- (61) a. John is eager PRO to please
 b. John is easy to please ____.

(Chomsky, 1964, 34)

Again, there is some room for variation. In some languages, again like French (Authier, 1989), only a true direct object can move. Adjuncts and indirect objects cannot be gapped. English (and Thai) are somewhat freer in allowing the gap to be any non-subject position that is in principle targetable for movement.²³

The lack of a subject gap remains largely unexplained as a general property of the *tough*-construction. After all, all the languages discussed have subject gaps in nonfinite clauses associated with e.g., *too/enough*-clauses. While there are plausible reasons for the lack of subject gaps on a language-by-language basis (for example, some languages employ nonfinite clauses that lack a subject position, as noted above), the fact that there is *no* language that allows a subject gap is puzzling, particularly since there are clearly languages that have nonfinite clauses which can include an overt subject, like Thai.

We put these four properties—thematic displacement, nonfiniteness, movement, and non-subject gaps—as the characterizing properties for all *tough*-constructions, across all languages and language families (that have a *tough*-construction). They constitute the “*tough*-signature.” Apparent exceptions to these properties—as Thai appears to exhibit—are due to language specific properties that are independent of the *tough*-construction. In the case of Thai, like in Japanese, and Arabic, there is the possibility of a broad subject position. Importantly, why these four properties are the building blocks for all *tough*-constructions is unknown. What is it about the confluence of these properties that allow *tough*-constructions?

We have suggested above an explanation for Thai which relies on the idea of complex-predicate formation. Extending this idea, we might attribute the characteristics of the *tough*-construction—including the crosslinguistic variation—to how languages accomplish this kind of complex-predicate formation. Assuming that a *tough*-predicate needs to combine with a particular kind of clause to create a property of individuals, the variation we see across the world’s language stems from the language-internal properties of these clauses.

References

- Alexiadou, A. and Anagnostopoulou, E. (2020). A comparative study of English and Greek *tough*-movement constructions. *Languages*, 2(218):17–38.
- Alexopoulou, T., Doron, E., and Heycock, C. (2004). Broad subjects and clitic left dislocation. In Adger, D., de Cat, C., and Tsoulas, G., editors, *Peripheries: Syntactic edges and their effects*, pages 329–358. Springer Netherlands, Dordrecht.
- Authier, J.-M. (1989). Two types of empty operator. *Linguistic Inquiry*, 20(1):117–125.
- Authier, J. M. and Reed, L. (2008). *Tough*-movement and nominalized infinitives in french. *Kansas Working Papers in Linguistics*, 30:1–12.
- Berman, A. and Szamosi, M. (1972). Observations on sentential stress. *Language*, 48(2):304–325.
- Browning, M. (1987). *Null Operator Constructions*. PhD thesis, Massachusetts Institute of Technology.

²³We are also aware of some languages, specifically Austronesian languages, which are purported to have subject gaps in the *tough*-construction (cf. Seiter 1979 for Niuean and Polinsky 2016 for Tongan). It’s not clear if this is the same phenomenon, however, since so-called *tough*-construction are part of a larger paradigm of “raising” out of the embedded clauses. Still, we feel it is necessary to point out that there may be true exceptions to the “non-subject gap” criterion.

- Cardinaletti, A. (1990). *Impersonal Constructions and Sentential Arguments in German*. Unipress.
- Cardinaletti, A. (1997). Subjects and clause structure. In Haegeman, L., editor, *The new comparative syntax*. Longman.
- Cardinaletti, A. (2004). Toward a cartography of subject positions. In Rizzi, L., editor, *The structure of CP and IP. The cartography of syntactic structures*, pages 115–165. Oxford University Press.
- Cardona, L. R. and Villalba, X. (2025). The interaction between clause size and Voice: Evidence from Catalan and Italian. *Glossa: a journal of general linguistics*, 10(1):1–51.
- Chomsky, N. (1964). *Current Issues in Linguistic Theory*. Mouton, The Hague.
- Chomsky, N. (1977). On *Wh*-Movement. In Culicover, P., Wasow, T., and Akmajian, A., editors, *Formal Syntax*. Academic Press, New York.
- Comrie, B. (1997). Tough-movement and its analogues in Germanic languages. In *Language and its Ecology*, pages 303–321. De Gruyter.
- Comrie, B. and Matthews, S. (1990). Prolegomena to a typology of Tough Movement. In Croft, W., Denning, K., and Kemmer, S., editors, *Studies in Typology and Diachrony: Papers presented to Joseph H. Greenberg on his 75th birthday*, volume 20 of *Typological Studies in Language*. John Benjamins.
- Dalrymple, M. and King, T. H. (2000). Missing-object constructions: Lexical and constructional variation. In Butt, M. and King, T. H., editors, *Proceedings of the LFG00 Conference*, Stanford, CA. CSLI Publications.
- den Dikken, M. and Singhapreecha, P. (2004). Complex noun phrases and linkers. *Syntax*, 7(1):1–54.
- Doron, E. and Heycock, C. (1999). Filling and licensing multiple specifiers. In Adger, D., Pintzuk, S., Plunkett, B., and Tsoulas, G., editors, *Specifiers: Minimalist Approaches*, pages 69–89. Oxford University Press, Oxford.
- Epstein, S. (1989). Quantification in null operator constructions. *Linguistic Inquiry*, 20:647–658.
- Fleisher, N. (2013). On the Absence of Scope Reconstruction in *Tough*-Subject A-Chains. *Linguistic Inquiry*, 44(2):321–332.
- Giurgea, I. and Soare, E. (2020). Agree of not: Tough choice. a study of *tough*-constructions in Romanian in a comparative perspective. *Langages*, 2(218):39–51.
- Gluckman, J. (2021). The Meaning of the *Tough*-Construction. *Natural Language Semantics*, 29:453–499.
- Gluckman, J. (2022). Taking time with the tough-construction. *Journal of Linguistics*, 58(2):233–268.
- Gluckman, J. (2024). Proleptic Objects as Complex-NPs. *The Linguistic Review*, 41:765–798.
- Hattori, R. (2023). Size of Op in *tough*-constructions. In Shen, Z. and Laszakovits, S., editors, *The Size of Things II*, pages 21–42. Language Science Press, Berlin.
- Havu, J. and Höglund, M. (2015). A cross-linguistic perspective on complementation in the *Tough* construction. In Höglund, M., Rickman, P., Rudanko, J., and Havu, J., editors, *Perspectives on Complementation: Structure, Variation, and Boundaries*, pages 52–76. Palgrave Macmillan.

- Heycock, C. (1994). *Layers of predication: The non-lexical status of clauses*. Garland, New York.
- Heycock, C. and Doron, E. (2003). Categorical subjects. *Gengo Kenkyu*, 123:95–135.
- Hicks, G. (2003). “*So Easy to Look at, So Hard to Define*”: *Tough Movement in the Minimalist Framework*. PhD thesis, University of York.
- Hicks, G. (2009). *Tough* Constructions and their Derivations. *Linguistic Inquiry*, 40(4):535–566.
- Holl, D. (2010). *Modale Infinitive und dispositionelle Modalitat im Deutschen*. Akademie Verlag, Berlin.
- Hoonchamlong, Y. (1991). *Some issues in Thai anaphora: A government and binding approach*. The University of Wisconsin-Madison.
- Inoue, K. (1978). ‘tough’ sentences in Japanese. In Hinds, J. and Howard, I., editors, *Problems in Japanese syntax and semantics*, pages 122–154. Kaitakusha.
- Inoue, K. (2004). Japanese ‘tough’ sentences revisited. *Scientific approaches to language*, 3:75–111.
- Iwasaki, S. (2002). Proprioceptive-state expressions in Thai. *Studies in Language. International Journal sponsored by the Foundation Foundations of Language*, 26(1):33–66.
- Iwasaki, S. and Ingkaphirom, P. (2005). *A reference grammar of Thai*. Cambridge University Press.
- Jenks, P. (2011). *The Hidden Structure of Thai Noun Phrases*. PhD thesis, Harvard University.
- Jenks, P. (2014). Generalized clausal modifiers in Thai noun phrases. *Syntax*, 1(1):299–342.
- Kaenmuang, J., Piyamahapong, P., and Pittayaporn, P. (2025). *Mān* on the referentiality continuum of Thai. *Journal of the Southeast Asian Linguistics Society (JSEALS)*, 18(1):1–18.
- Keine, S. and Poole, E. (2017). Intervention in *tough*-constructions revisited. *The Linguistic Review*, 34(2):295–329.
- Kim, B. (1995). *A Causative Analysis of Tough-Constructions*. PhD thesis, University of Texas at Austin.
- Klingvall, E. (2018). Agreement and reconstruction correlate in Swedish. *Natural Language & Linguistic Theory*, 36(4):1165–1205.
- Kuno, S. (1973). *The structure of Japanese*. MIT Press, Cambridge, MA.
- Kusayama, M. (1998). Middle and *Tough*-constructions in English. *Tsukuba English Studies*, 17:201–219.
- Landau, I. (2011). Predication vs. aboutness in copy raising. *Natural Language & Linguistics Theory*, 29(3):779–813.
- Lasnik, H. and Fiengo, R. (1974). Complement Object Deletion. *Linguistic Inquiry*, 5(4):535–571.
- Levine, R. and Hukari, T. E. (2006). *The Unity of Unbounded Dependency Constructions*. CSLI Publications, Stanford, CA.
- Massam, D. (1987). Middles, tough, and recipe context constructions in English. In *Northeastern Linguistic Society*, volume 18, pages 315–332.

- Meechanyakul, N. (2024). On the binding preference for Thai reflexives: The role of morphology, syntax, and pragmatics. *Journal of the Southeast Asian Linguistics Society*, 17(2):1–17.
- Ohkado, K. (1993). *Tough constructions in Japanese*. Master’s thesis, McGill University.
- Piriyawiboon, N. (2011). *Classifiers and determiner-less languages: The case of Thai*. PhD thesis, University of Toronto.
- Polinsky, M. (2016). *Deconstructing Ergativity: Two Types of Ergative Languages and Their Features*. Oxford University Press, Oxford.
- Poole, E., Keine, S., and Mendia, J. A. (2017). More on (the lack of) reconstruction in *tough*-constructions. lingbuzz/003702.
- Portner, P. and Yabushita, K. (1998). The semantics and pragmatic of topic phrases. *Linguistics and Philosophy*, 2:117–157.
- Rawlins, K. (2013). About *about*. In *Proceedings of SALT 23*, pages 336–357.
- Rezac, M. (2006). On Tough Movement. In Boeckx, C., editor, *Minimalist Essays*. John Benjamins, Amsterdam/Philadelphia.
- Ruangjaroon, S. (2005). *The Syntax of Wh-Expressions as Variables in Thai*. PhD thesis, University of British Columbia.
- Saito, M. (1985). *Some Asymmetries in Japanese and their Theoretical Implications*. PhD thesis, Massachusetts Institute of Technology.
- Seiter, J. W. (1979). *Studies in Niuean Syntax*. PhD thesis, University of California, San Diego.
- Singhapreecha, P. (2010). Modals and control infinitives: A view of Thai. *Lingua*, 120(5):1288–1312.
- Singhapreecha, P. (2019). Non-finite clauses in Thai. *Journal of Southeast Asian Linguistics Society*, 12(1):25–51.
- Stowell, T. (1991). The alignment of arguments in adjectives phrases. In Rothstein, S., editor, *Perspectives on Phrase Structure: Heads and Licensing*, Syntax and Semantics 25, pages 105–138. Academic Press, San Diego.
- Sudmuk, C. (2003). The *thuuk* constructions in Thai. In Butt, M. and King, T. H., editors, *Proceedings of the LFG 03 Conference*, pages 402–423.
- Takezawa, K. (1987). *A configurational approach to case-marking in Japanese*. PhD thesis, University of Washington.
- Takita, K., Maeda, M., and Nakamura, T. (2022). Varieties of tough-constructions and FormCopy. In *Proceedings of the 13th Generative Linguistics in the Old World in Asia (GLOW in Asia XIII)*, pages 257–269.
- Tayalati, F. and Danckaert, L. (2020). The syntax and semantics of Modern Standard Arabic resumptive *tough*-constructions. *Folia Linguistica*, 54(1):197–238.

Tayalati, F., Mostrov, V., and Van de Velde, D., editors (2020). *Les construction Tough, syntax, sémantique et interfaces*, *Langages*, volume 218.

Villa-García, J. (2023). Hanging Topic Left Dislocations as extrasentential constituents: towards a paratactic account. Evidence from English and Spanish. *The Linguistic Review*, 40(2):265–310.